

How to design ERNIE rings

Step 1

Install the software. During this step, one could install either Autodesk Inventor/ Solid Works/ AutoCAD for CAD design.

<https://www.autodesk.com/education/edu-software/overview#INVPROSA>

Select the student version if you are a student to get a free version.

Step 2.

Navigate to the File of results and identify which ring you will design first. For example, ring 288. The inner bore is 80mm, and the outer bore is 160mm. Then the outermost circle, where the shim trays are added, is 190mm.

Configuration_Number	Band_Number	Band_Radii_Gap_mm	Magnet_Space_mm	Band_Separation_mm	Band_1_Radius_mm	Band_1_Magnet_Count
Ring_321	1	0	8.8235	16.9492	65.43415254237 29	16
Ring_288	1	0	11.7647	15.2881	63.77313559322 04	14
Ring_251	1	0	8.8235	13.6271	62.11211864406 78	15
Ring_42	1	0	10.2941	3.6610	52.14601694915 26	12

Total number of magnet used = (Ring_321(16 magnets), Ring_288(14 magnets) 2, Ring_251(15 magnets)*2 and Ring_42(12 magnets)*2). = 16+28+30+24 = 98 magnets.

Step 3.

After looking at design parameters, open Autodesk Inventor. Select **File> New> Templates> Metric> (Standard)mm> Create**. This sets the dimensions from the default (Inches to mm).

The display of the background is usually dark, so one can also change this to sky by selecting (**File> Options> Colors> Sky**).

Then select Apply to apply these changes.

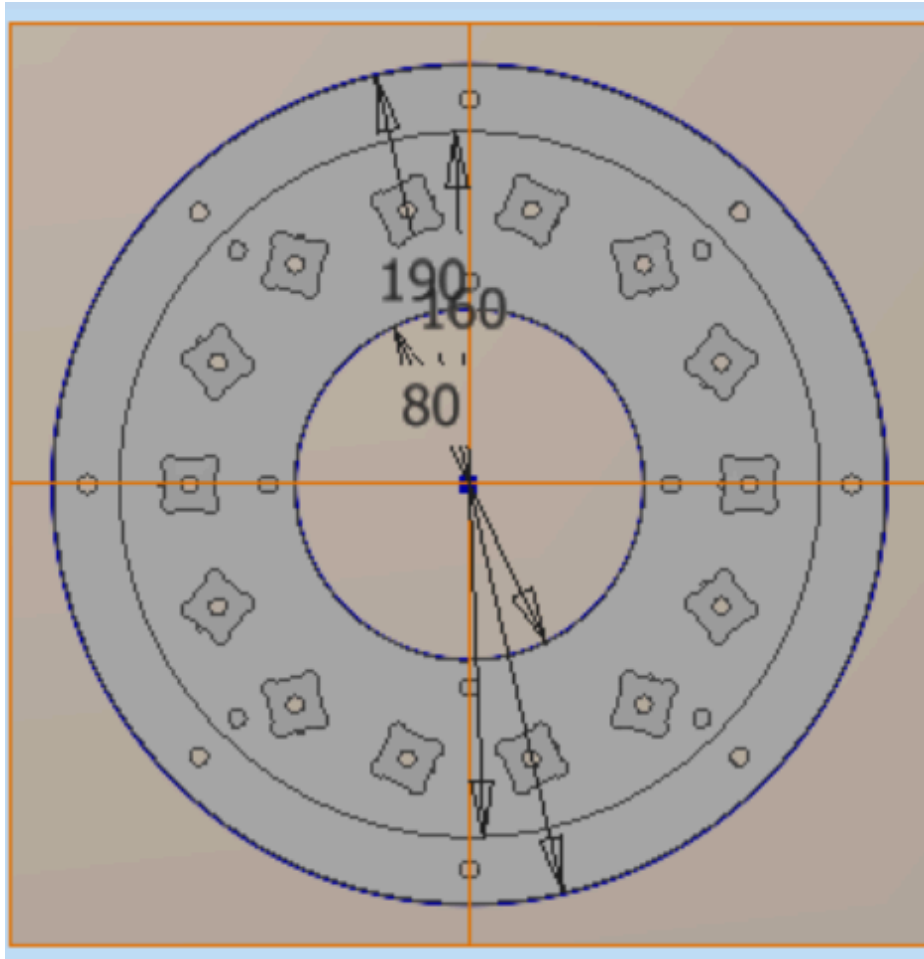
Step 4

When the new part is created, all the planes are invisible. To make them visible, select **Origin > Right-click Planes > Visible**. Do this for the YZ, XZ, and XY planes.

Right-click on the XY plane and make it perpendicular (or navigate to the **Sketch menu > select the perpendicular sign**).

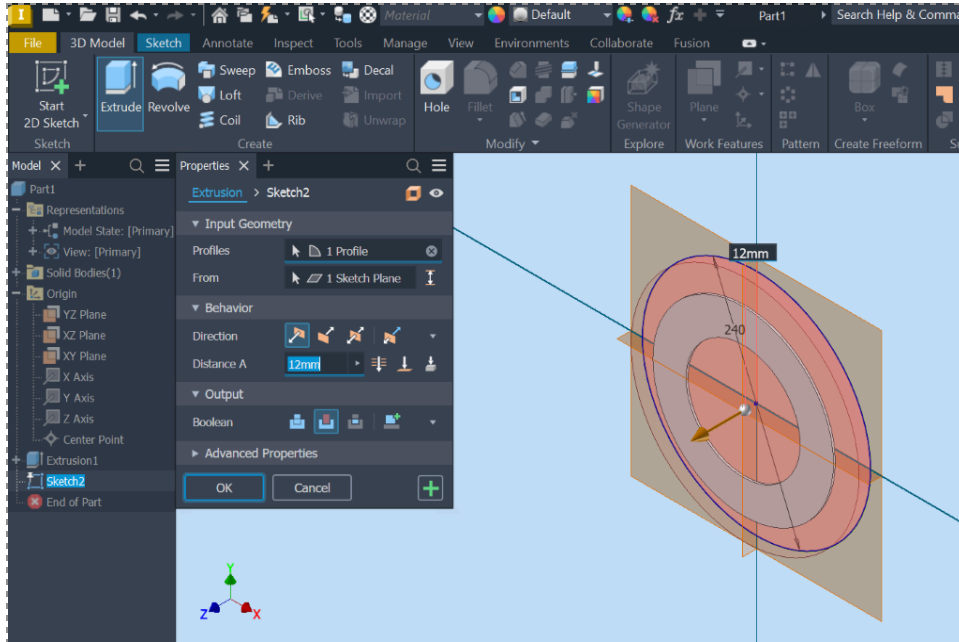
Step 5: Draw circles

To draw the inner bore, navigate to the **Sketch > Center circle**. Draw the circle while starting at the center. Set the diameter to 80mm. Draw a second circle for the outer bore of the ring and set its diameter as 160mm. Draw a third circle (for the shim trays) and set its diameter to 190mm.



Step 6: Extrude the circles

After drawing the circles, finish sketch. First extrude the inner circle by **14mm**. Finish the sketch. Then right click on the sketch and select visible. Then extrude the second face by **12mm**. Select the inner section and extrude it by 14mm. Select the outer region and extrude it by 12mm.

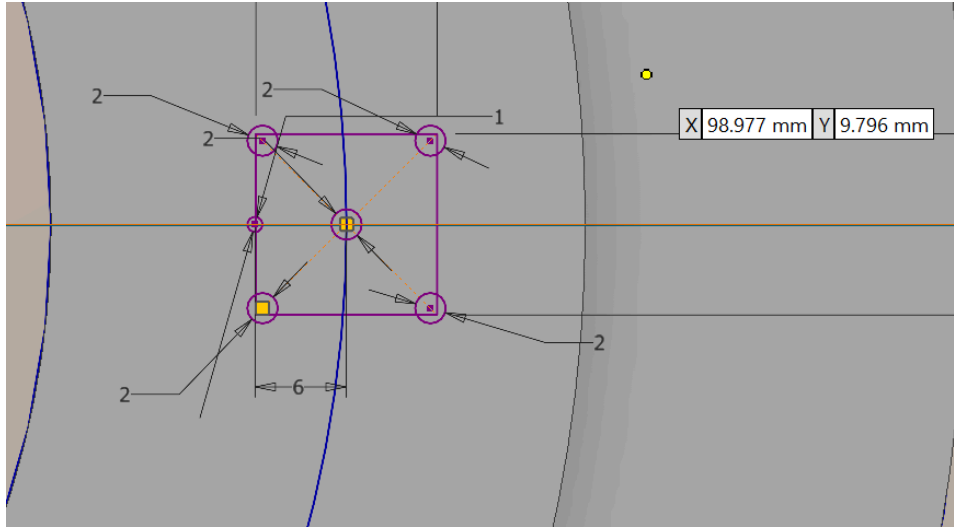


Step 7: Drawing the magnet Band

From the field results, the band radius is shown. For example, the Ring_288 has a band radius of 63.773mm. Set the entire diameter as 127.546mm. Press Enter. That is the band where the magnets will be.

Step 8: Create Magnet Block

Select Sketch 5, where the band-radius line is, and select Edit Sketch to draw the magnet block. Select the two side rectangles and draw a horizontal line that crosses the radius band circle at 180 degrees.



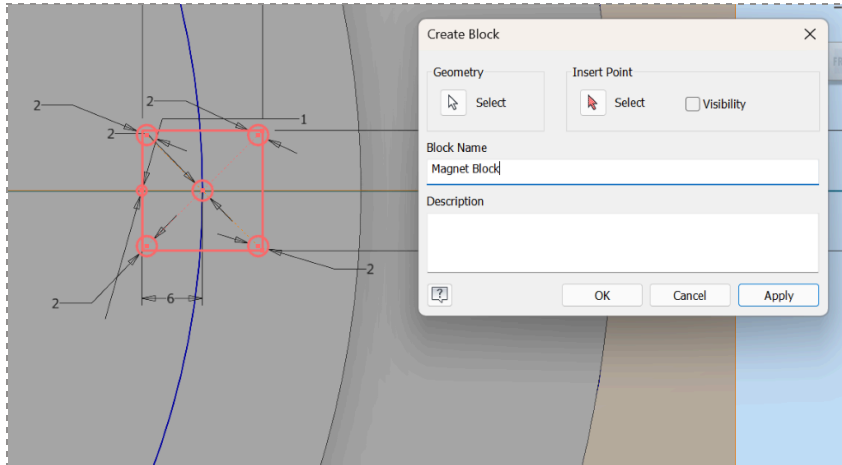
The dimensions of the magnet are 11.8 mm x 11.8 mm x 11.8 mm after measuring. So dimension the rectangle as 11.8mm. So it forms a square.

Draw a little dot on the side of the square facing the midpoint of the entire ring. This will give the direction in which the magnets will rotate. Dimension it such that the distance between the centre of the dot and the centre of the square is 6mm. Set the diameter of the dot to 1mm.

Draw other small circles at the four corners of the square. Set their diameters to 2mm each. There are such that the magnet can fit into the hole with ease.

Draw another circle at the center of the square and set its radius to 2mm. This is so as to ease the process of removing a magnet if necessary.

To create a magnet block, navigate to Folder Model> Block> Create Block. Select all the lines, circles, and points associated with the newly created block.

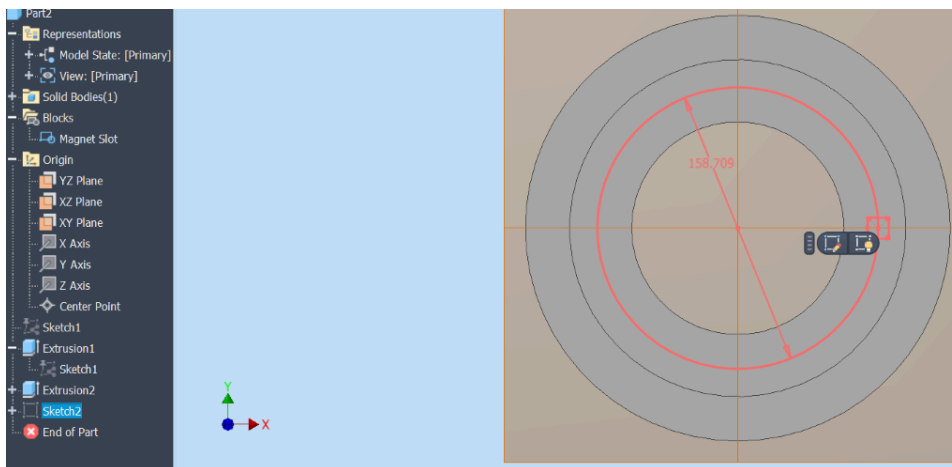


Select the insert point and choose the center of the block. This selects the point at which the block will be inserted at each point of the band-radius. Input a block Name as Magnet Block. Click on Apply to create it. All the center points of circles and rectangles should change to the color white.

Click the green tick to finish.

You can hide the other sketches to create more focus on the magnet block.

Then finish the sketch and hide it.



Still within the sketch mode, draw a new radius band circle and make its diameter 127.546.

Place a magnet block onto it.

If you try moving the block, it will move entirely. To fix it, draw a line from the center of the ring to its corresponding point on the radius band circle. Now place the magnet block at the point where this horizontal line meets the band. The center of the block should turn green.

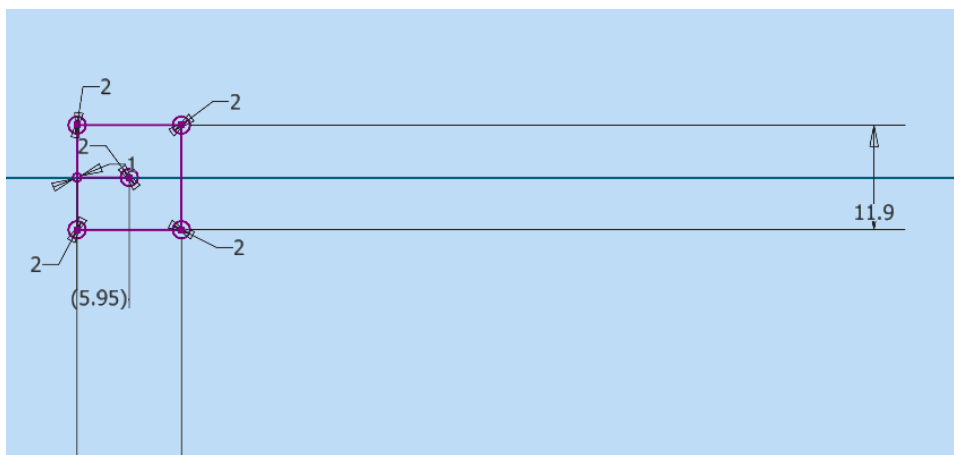
Try to move the block around, and it should rotate about the point at 360 degrees

From the results file, identify the number of magnets in the selected band for that specific ring.

For example, Ring_288 has 14 magnets.

Here, a pattern is made of the magnet blocks. Choose the circular pattern option, and select the horizontal line from the center of the ring to the center of the magnet block. Select the ring center as the axis and make 7 patterns. Select OK.

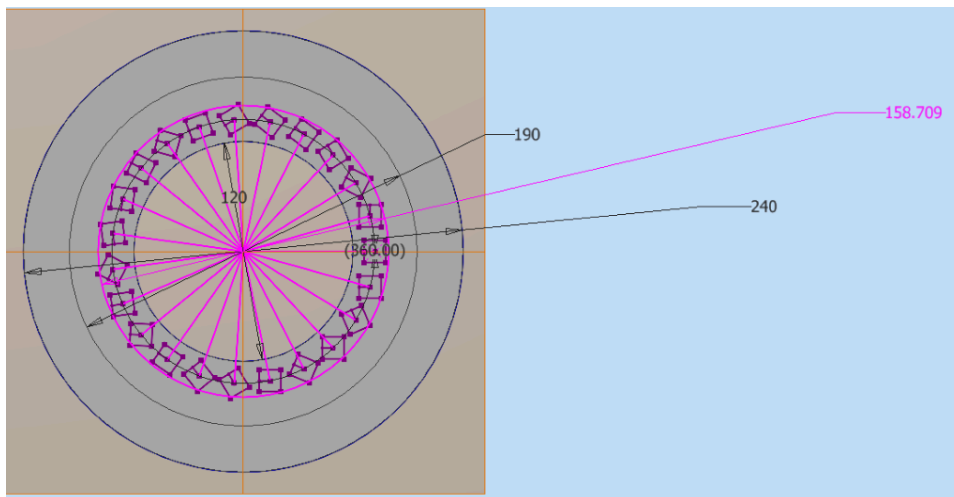
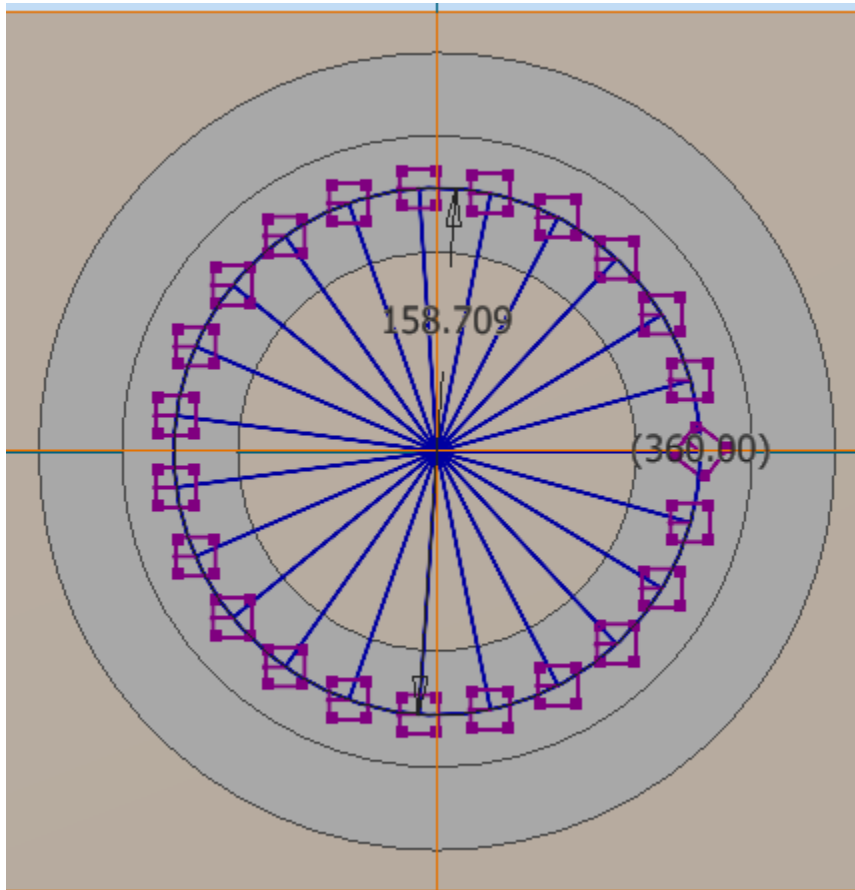
Select the magnet block from the menu and place it at every end of the line. Make sure the center point of the magnet (Insert point) turns green when the magnet block is placed at every point.



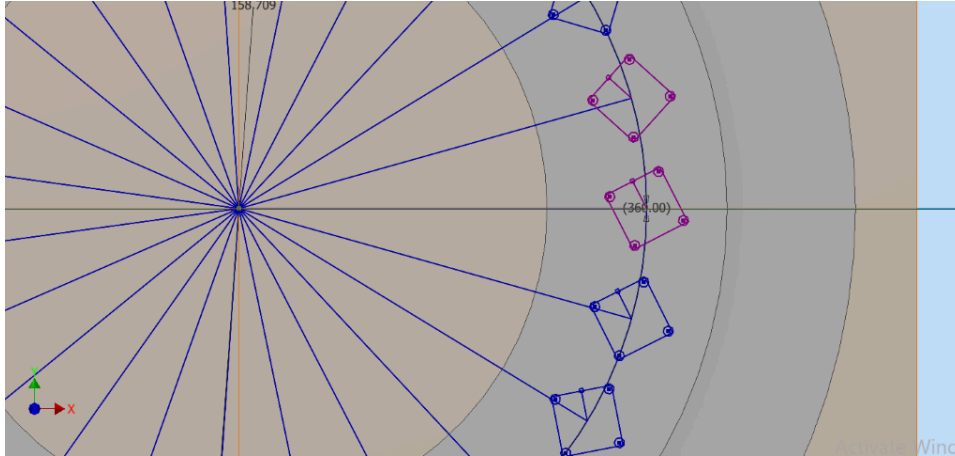
Then, still in the sketch mode, select edit block.

Draw a line from the center of the block to the center of the small circle placed 6mm away from the center of the magnet slot.

Step 9: Dimension the angles between the magnets.



In preparation for dimensioning the magnet slots, tilt them so that they do not auto-connect to anything. This is done to make sure the dimensions are angular and not linear.



Step 10: Dimensions for the magnet slots.

Note: To simplify the dimensioning process, make sure that all the magnet slots are purple in color, hence flexible. That way, one can change their dimensions and angle of orientation. If the magnet slots are blue, then they are already fixed.

Here, use the formula for halbach array, which is: $360(2)/n$. N is the number of magnets.

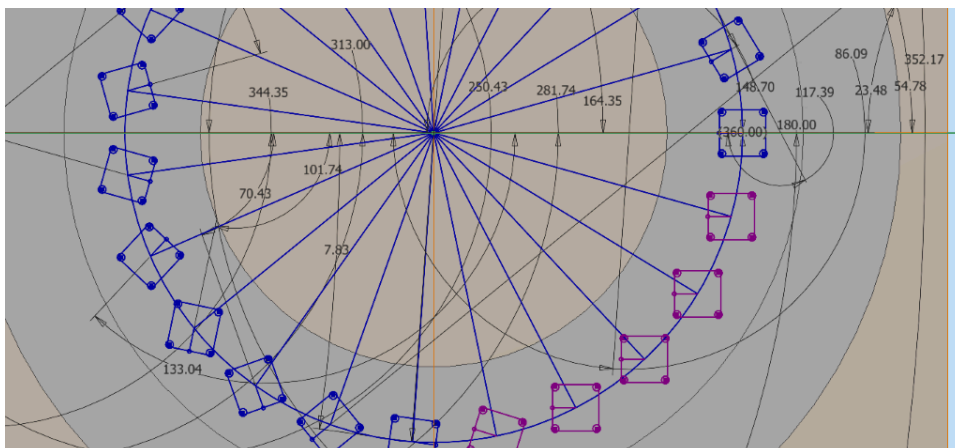
For example, for the 120mm inner bore diameter rat scanner, the first ring to be designed was the middle ring 842. From the results obtained from the algorithm, the ring had 20 magnets.

$$360(2)/20 = 36.$$

This means there is an angular difference of 36 degrees for each slot.

However, this value is negative because the magnet slots are rotated in the anti-clockwise direction.

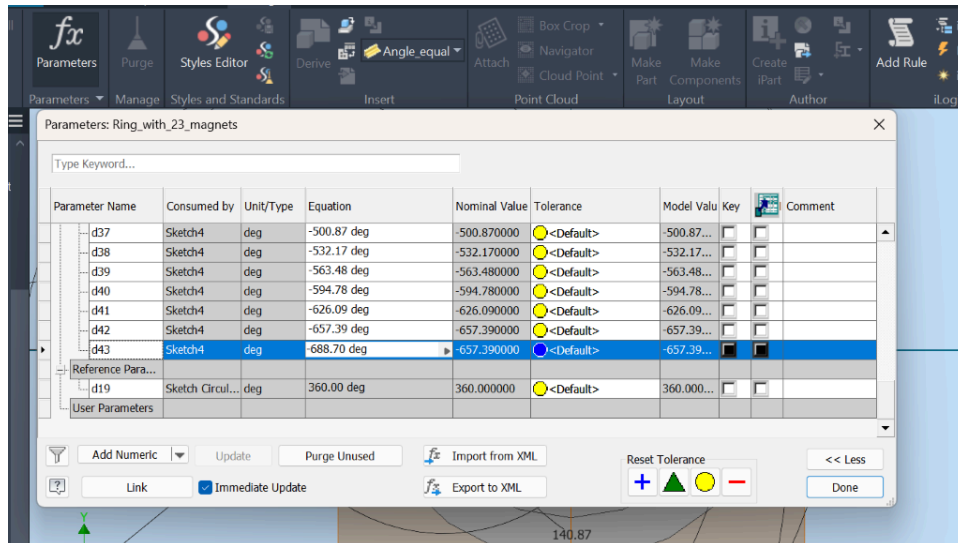
After setting the specific angular dimension for a slot, it attains a fixed orientation. The fixed slots are blue while the flexible ones are purple.



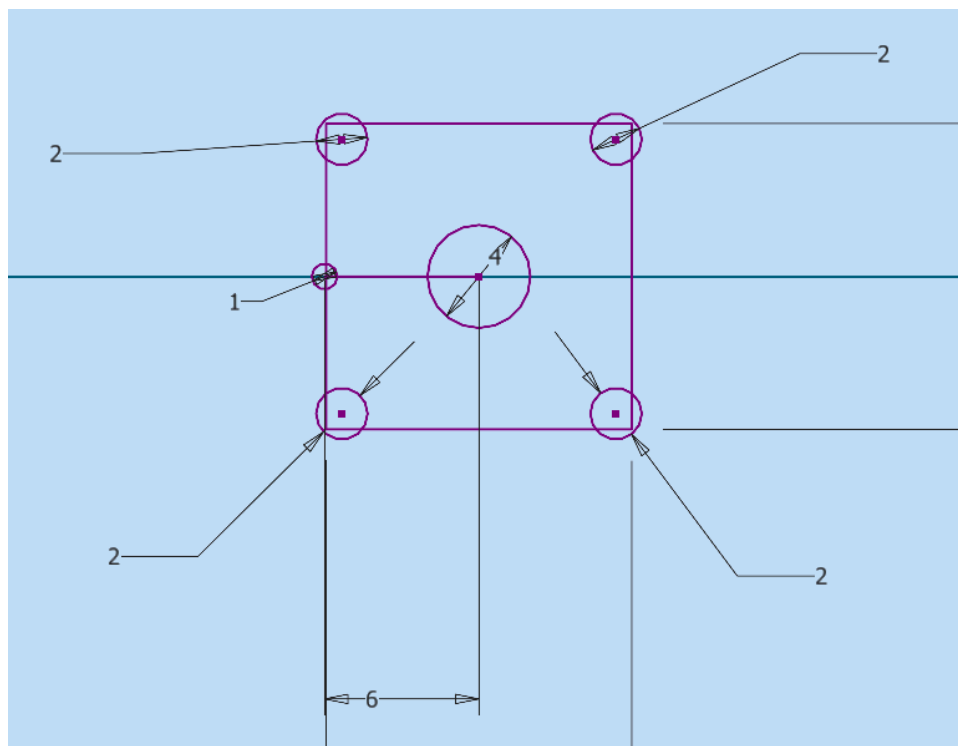
After putting in all the angular dimensions, all the rings should appear blue.

Next, navigate to **Manage> Parameters**. This enables one to look through all the angles and ensure that they are the right ones.

If there are any errors, correct them. Then, save.



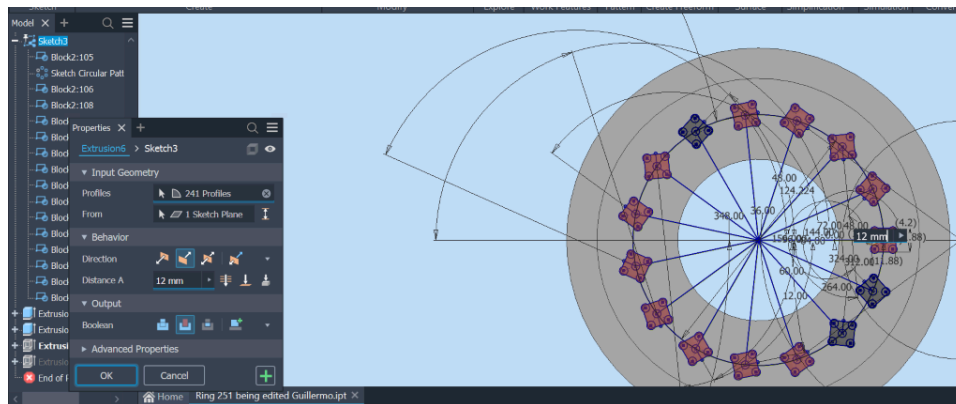
Edit the block and add a circle at the center for pushing out magnets



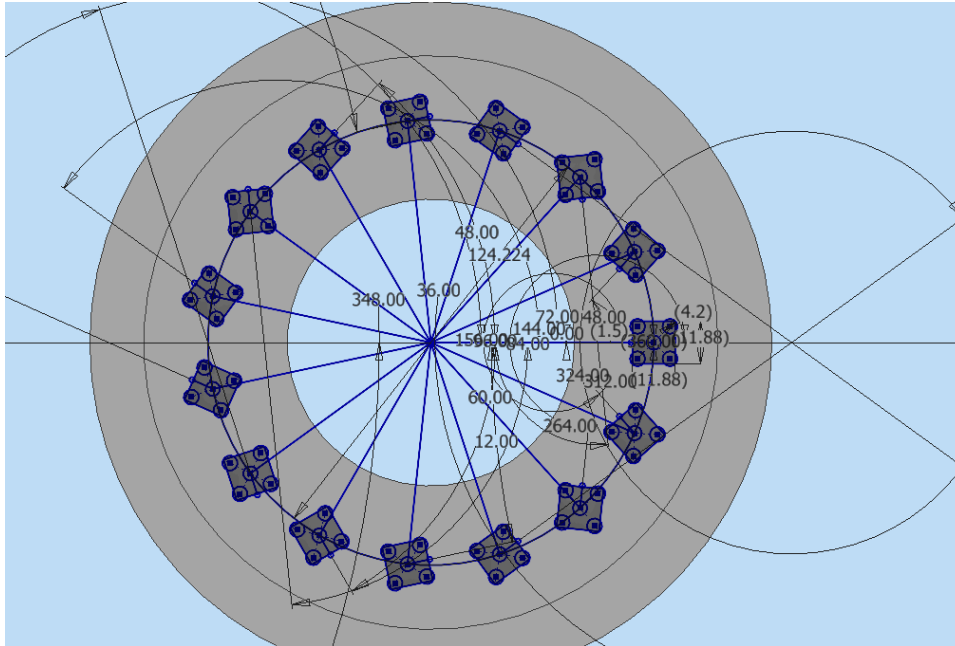
Since the magnets were measured as 11.88X 11.88X 11.88, these can be made as 12mm.

[illegible]

When all selected for extrusion

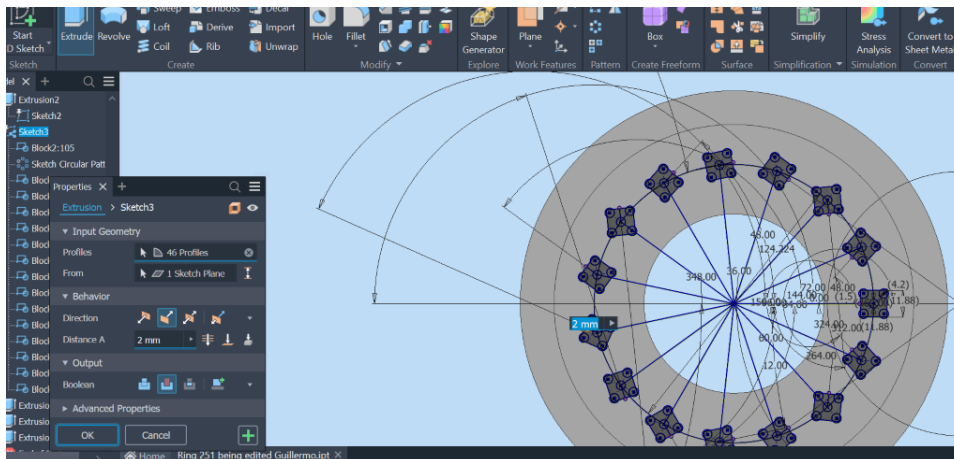


10

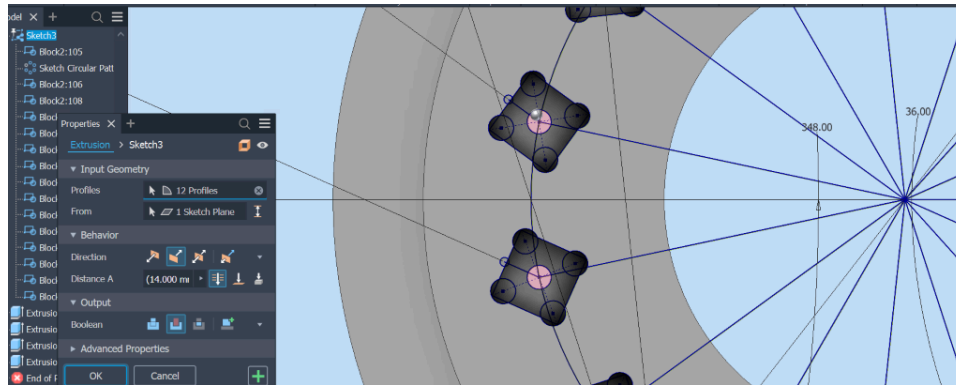


Now extrude the 1.5mm diameter holes.

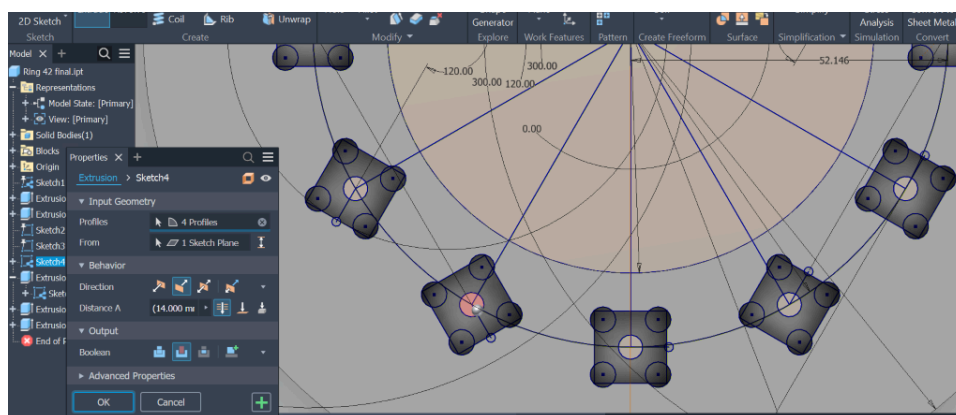
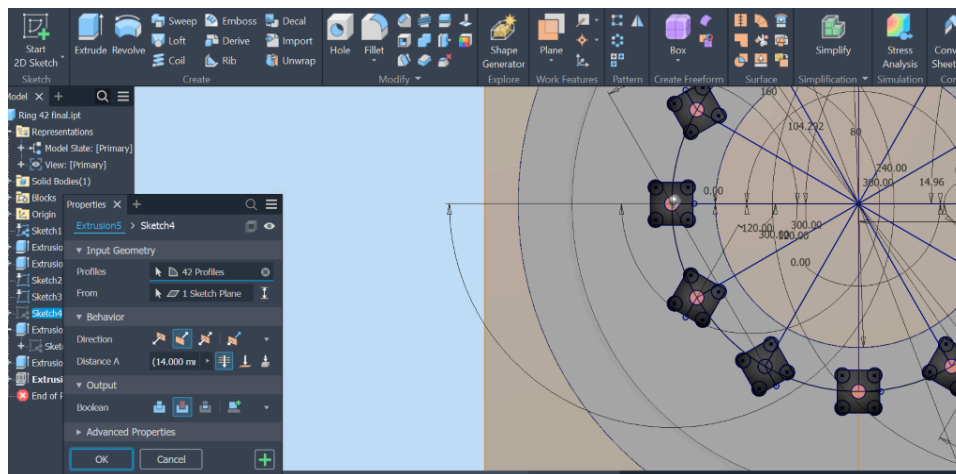
To extrude the small 1.5 diameter holes, select edit sketch, set the dimension of extrusion to 2mm and select all



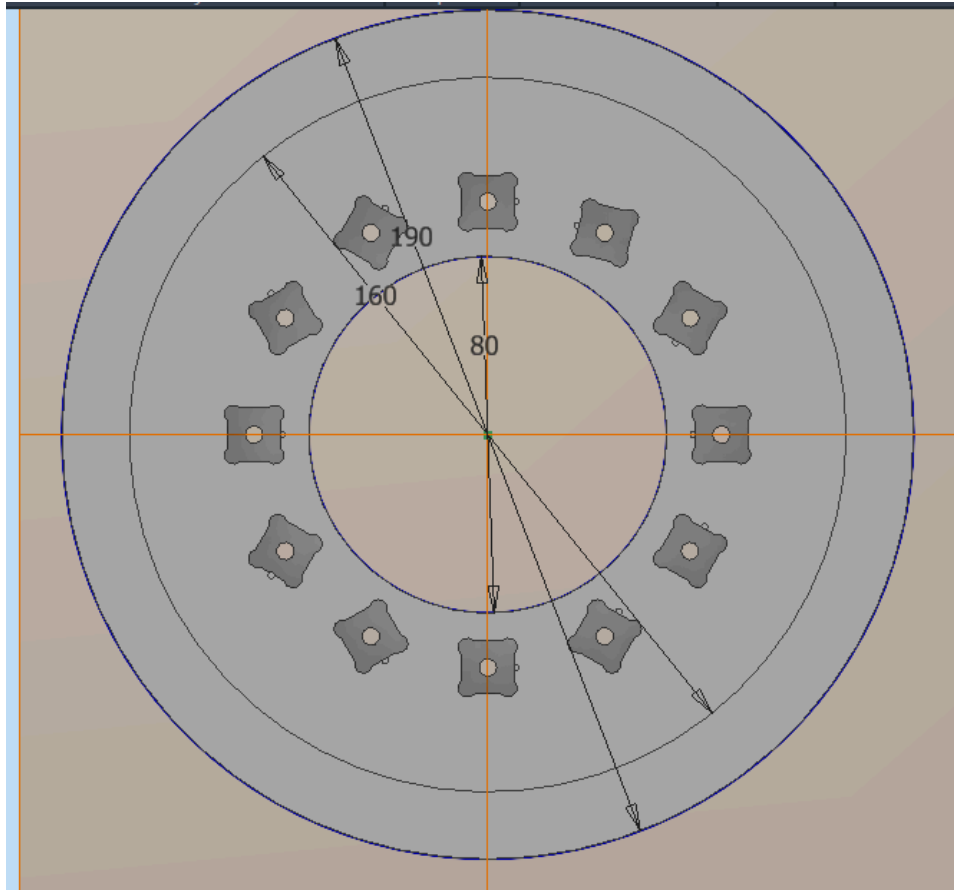
Next Step, extrude cut through all the 4mm holes



Cut through all



This is how the ring should look like



Note that the extrusion of where the magnets sit is by 12mm, and the hole for helping us to remove the magnet is through all extrude cut.

M3 = 4.2mm

M4 = 4.3mm

First, make the circles drawn for the hole extrusion visible.

Then, select extrude.

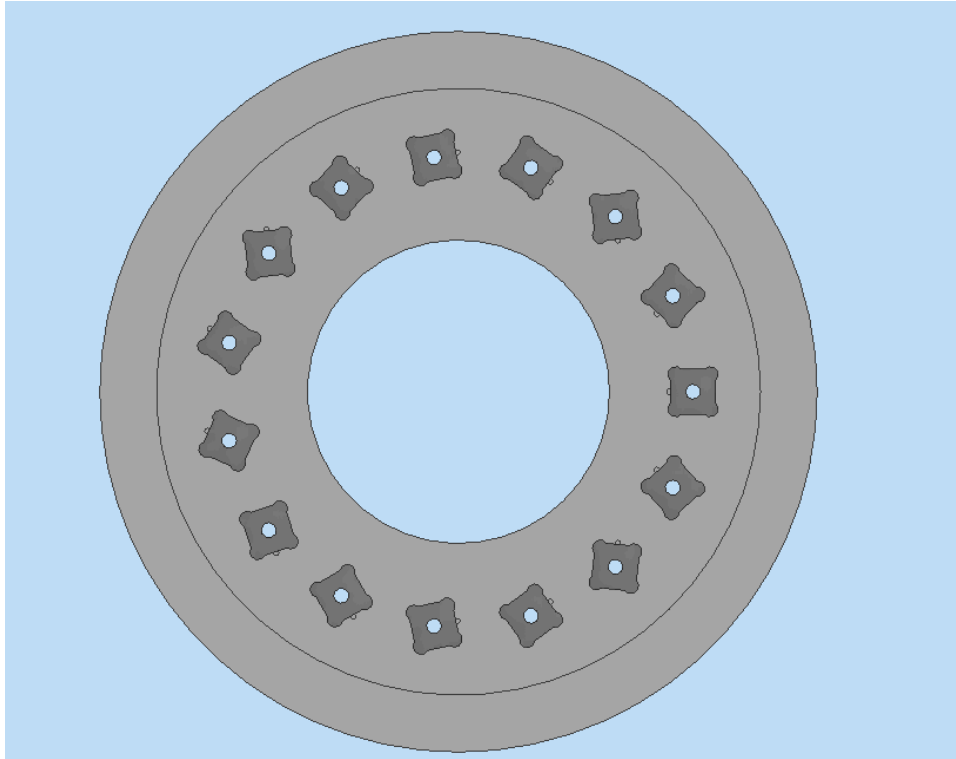
Later, draw the small circles for inserting the screws and lids.

After this action, draw the M4 circles where the brass rods will pass.

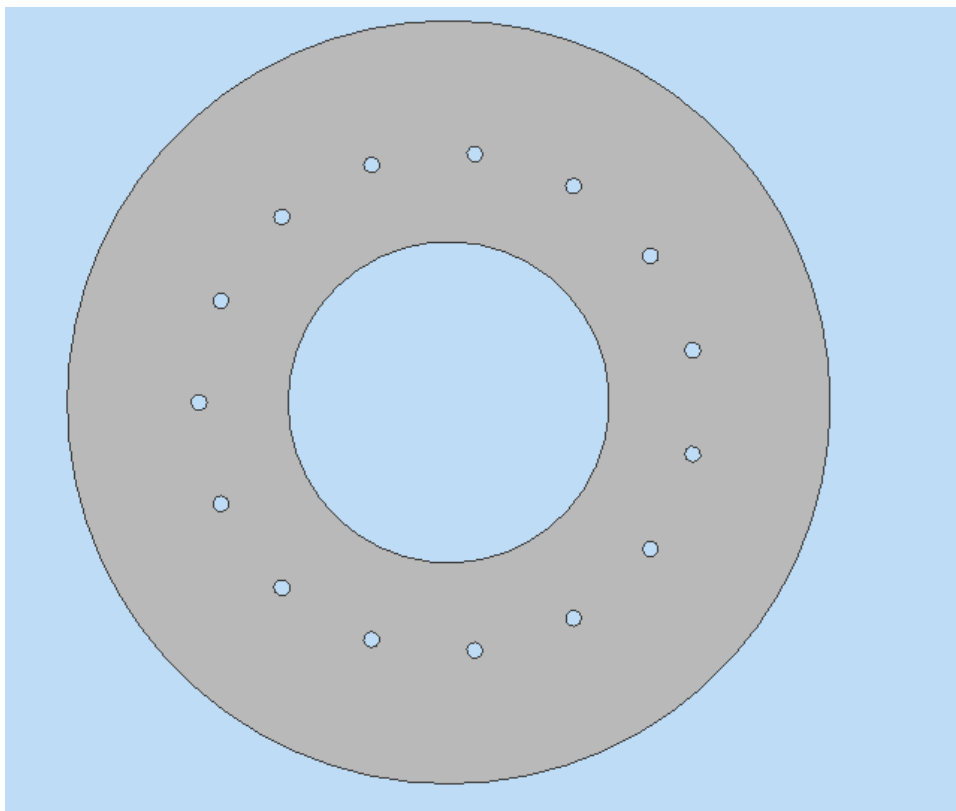
Extrude cut the Holes through all by 12mm

Here, make sketch visible, cut , through all

Ring 251 for 50mT small scanner final look before adding the other holes for the brass rods and screws that holed the lids tight onto the ring



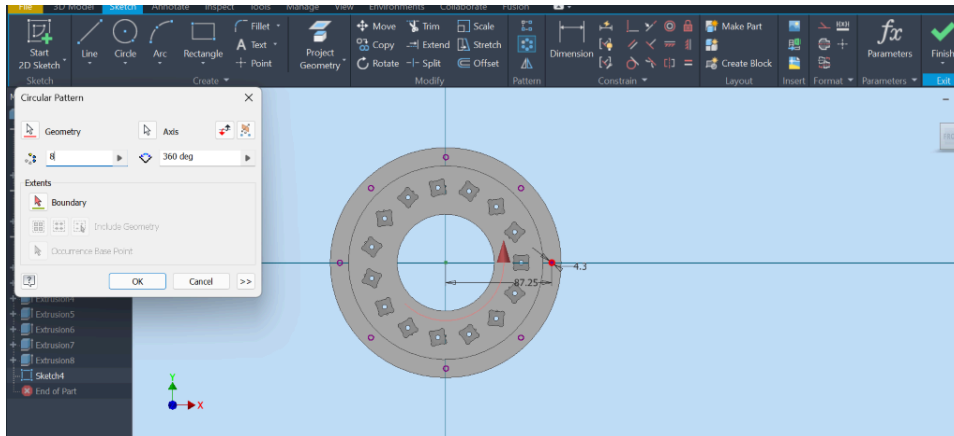
Back view



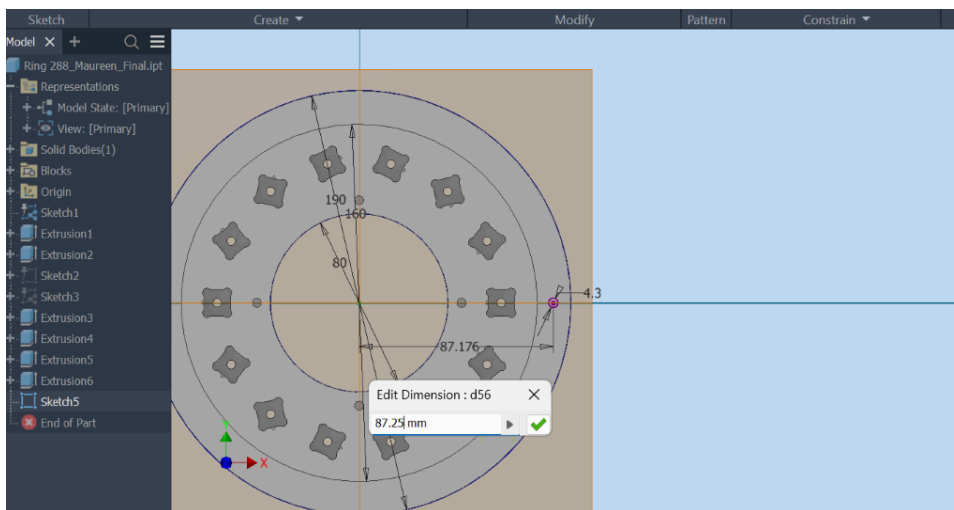
Draw a circle for the brass rod.

Make them 4.3mm each.

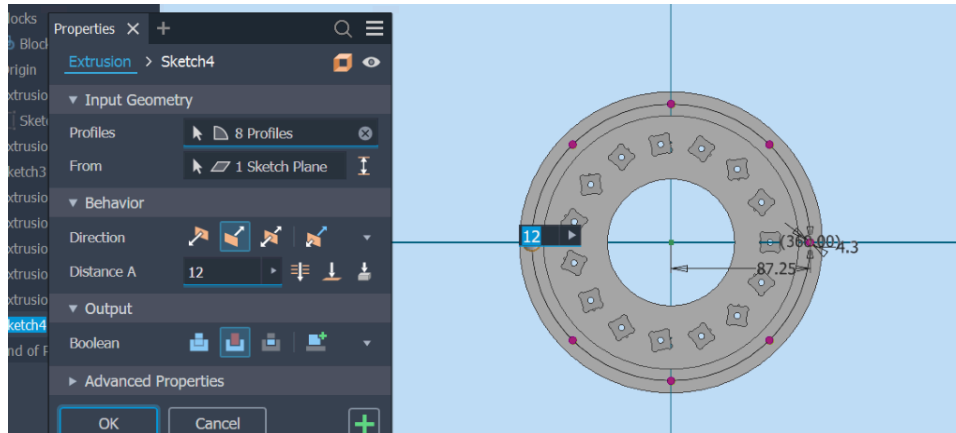
Then select edit sketch and create a circular pattern of the 4.3mm circle for 8 circles



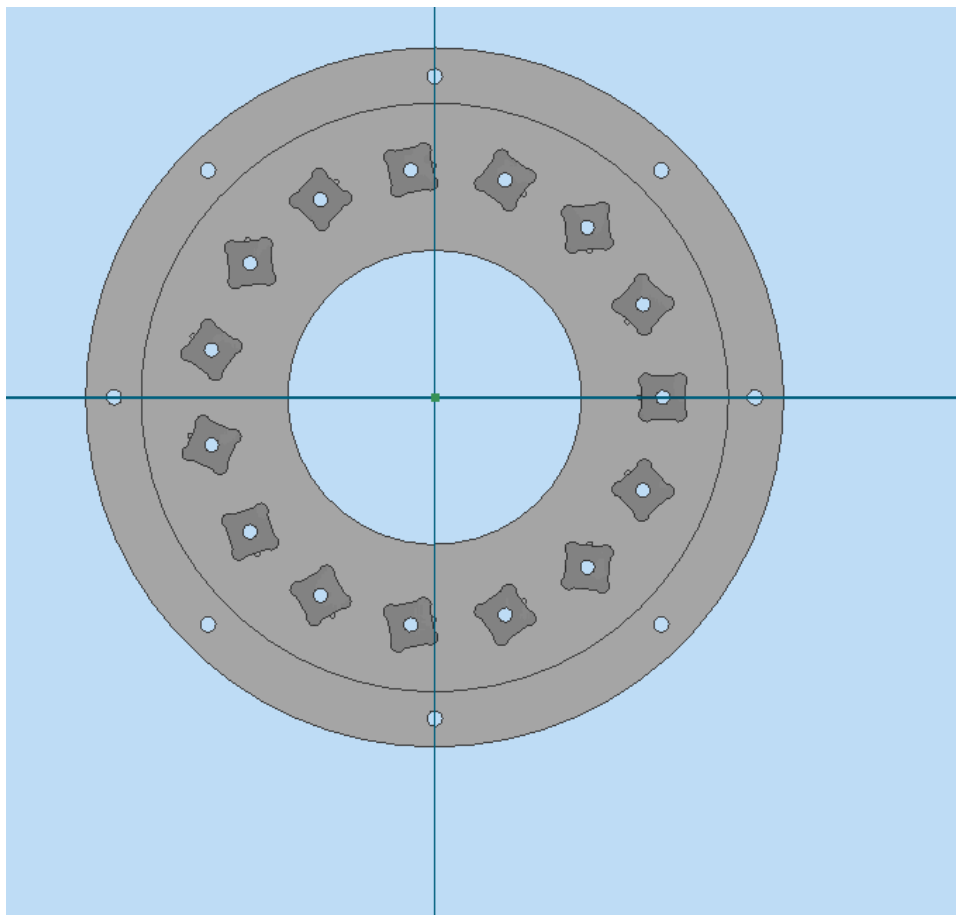
Extrude cut the holes (Always dimension the brass rod circles immediately after creating)



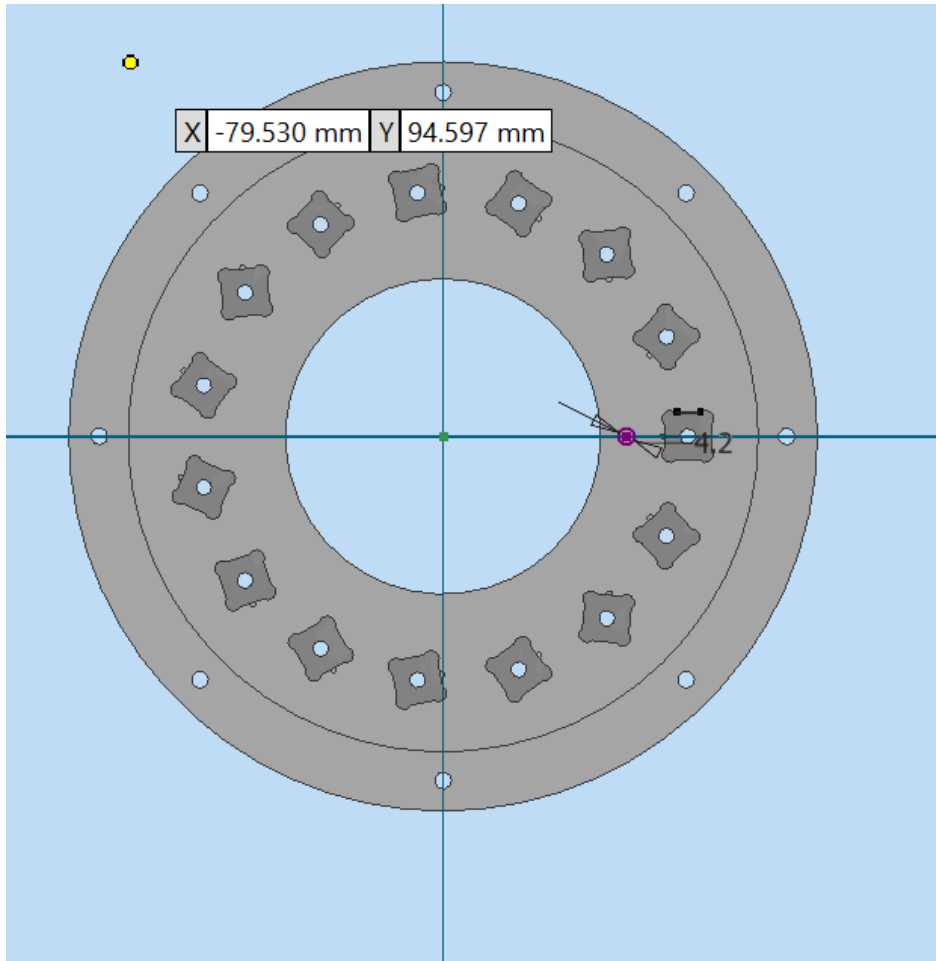
Re



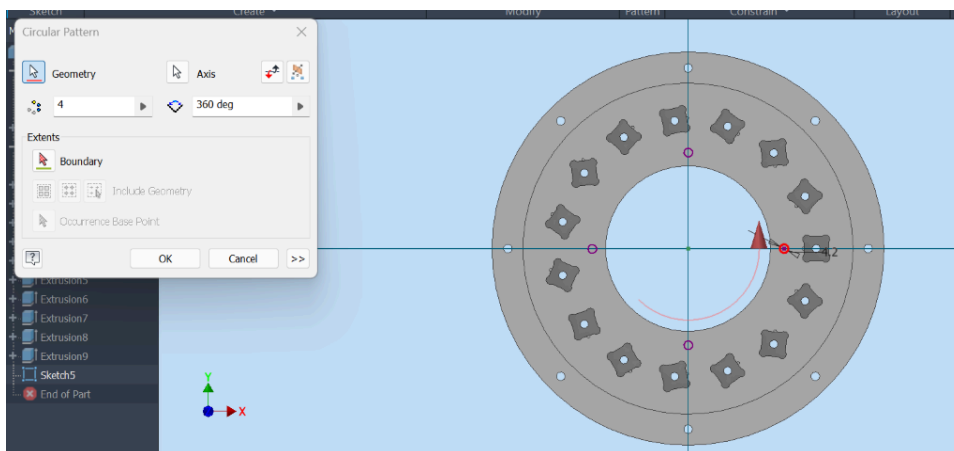
This is the final result



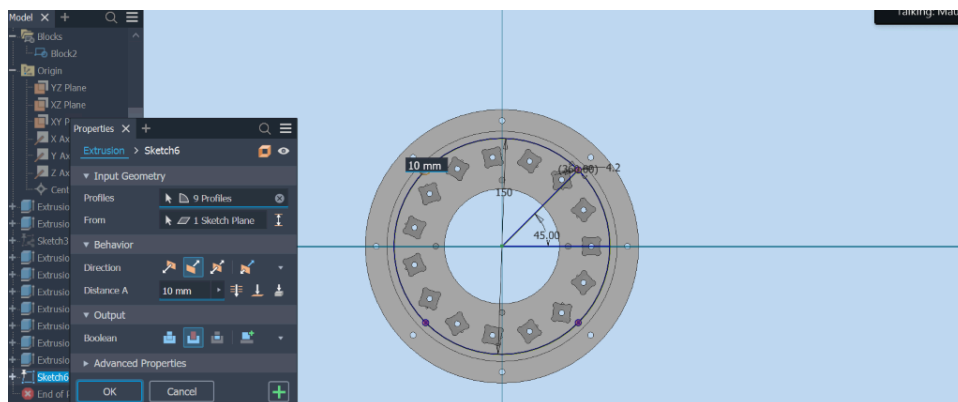
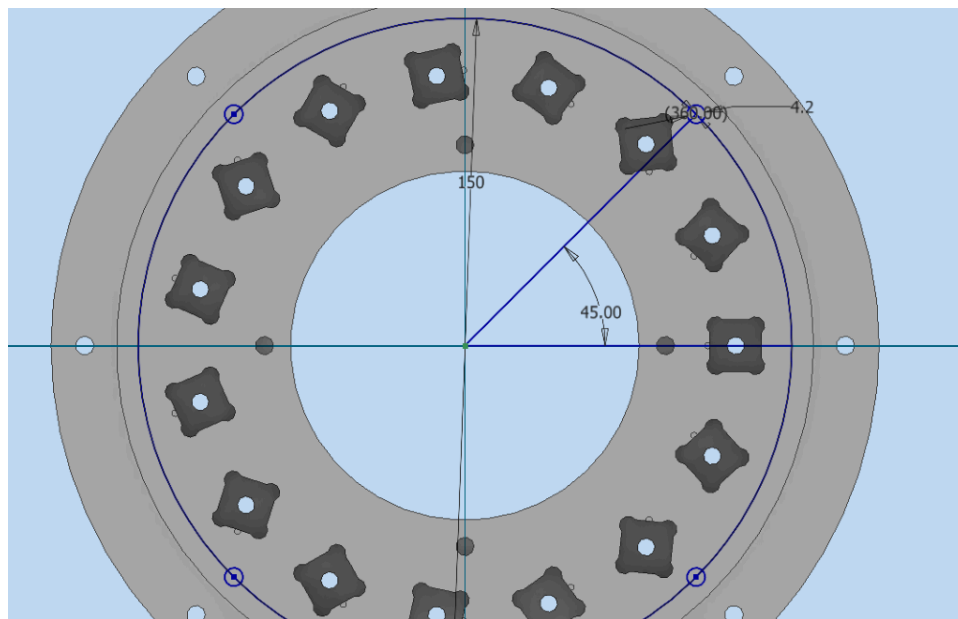
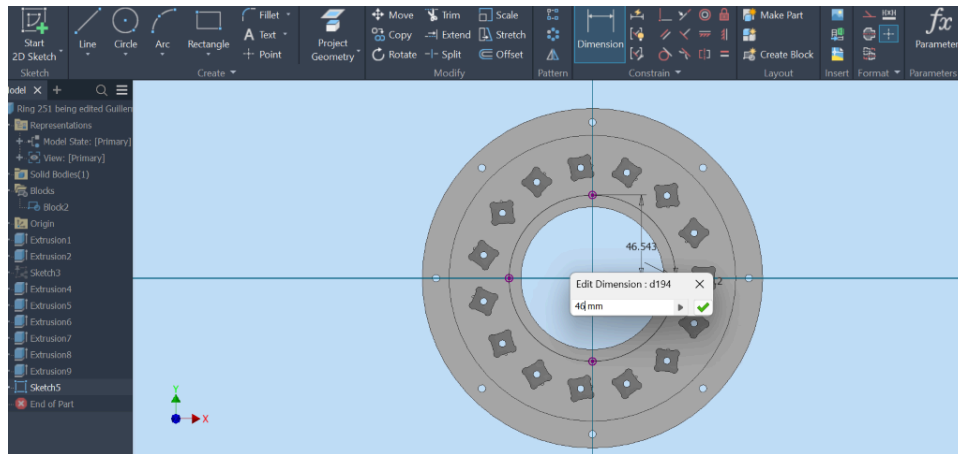
Now add circles on the interior ring area for the heat set inserts and screws for the lid
4.2 mm for M3



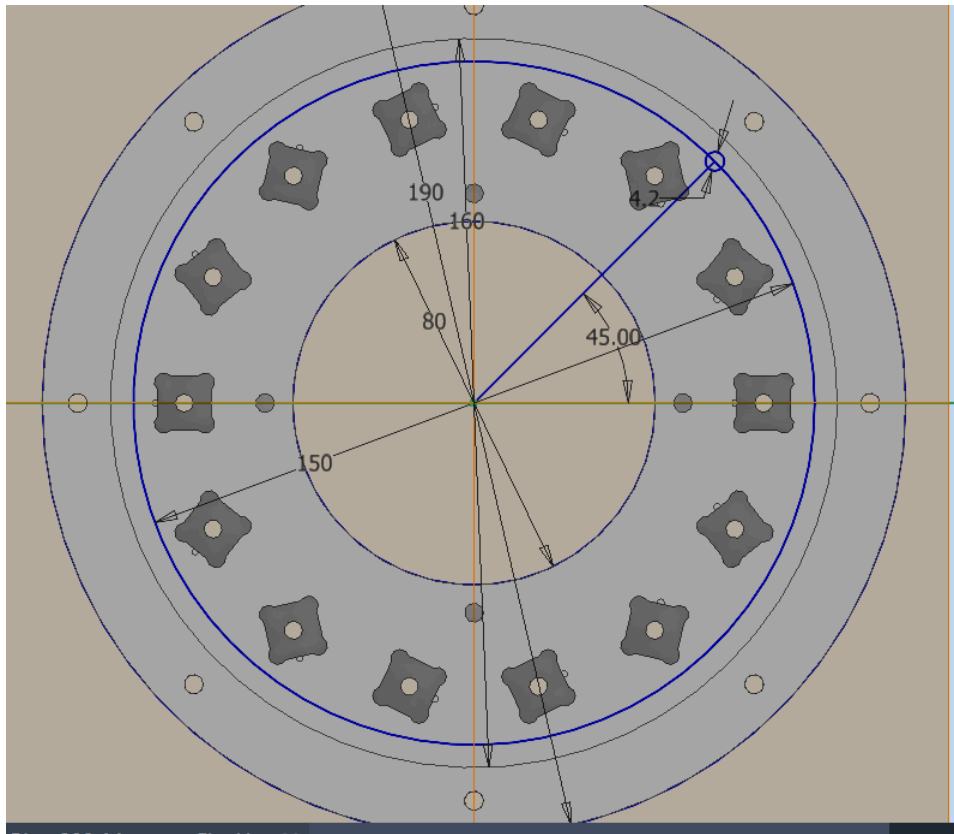
Result



Dimension



SECOND SET OF INNER CIRCLES FOR THE SCREWS

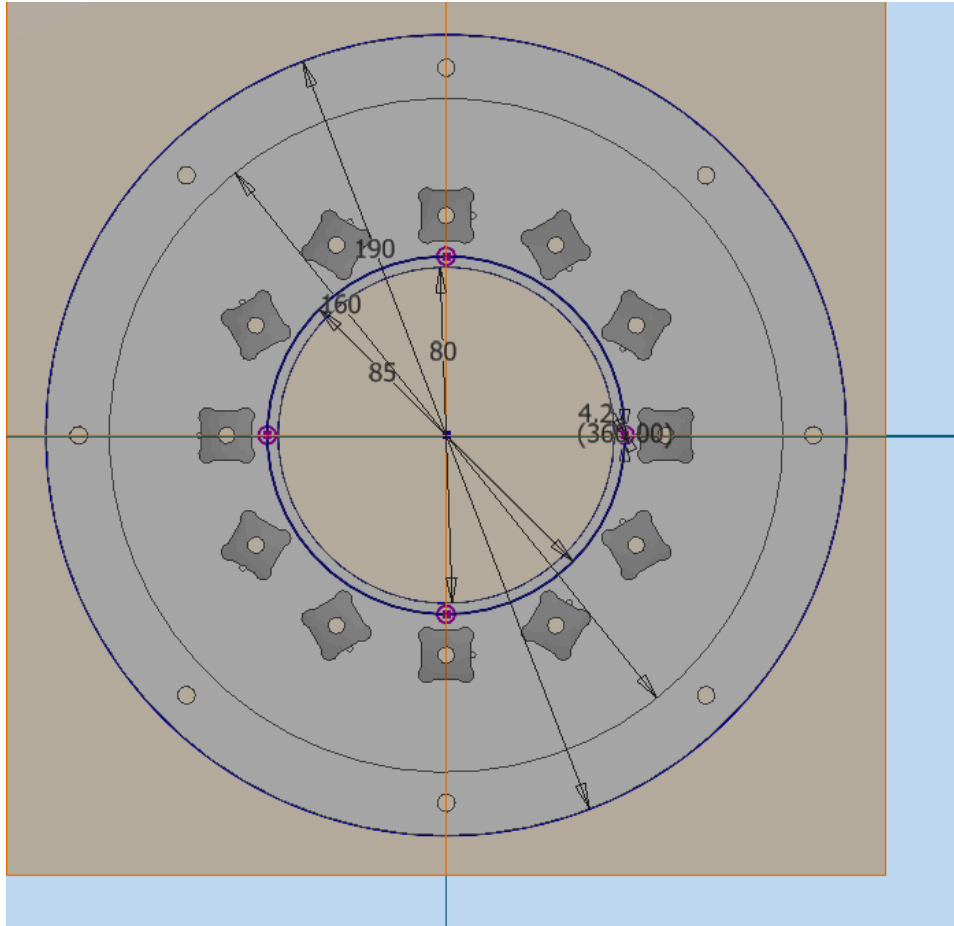


They should be at angle 45

First draw circle whose diameter is 150

Add a circle whose diameter is 4.2 for M3

NOTE: While creating holes for insertion of the heat set inserts and then screws to tighten lids onto the rings, keen attention should be taken so that the holes cannot overlap with the magnet slots.



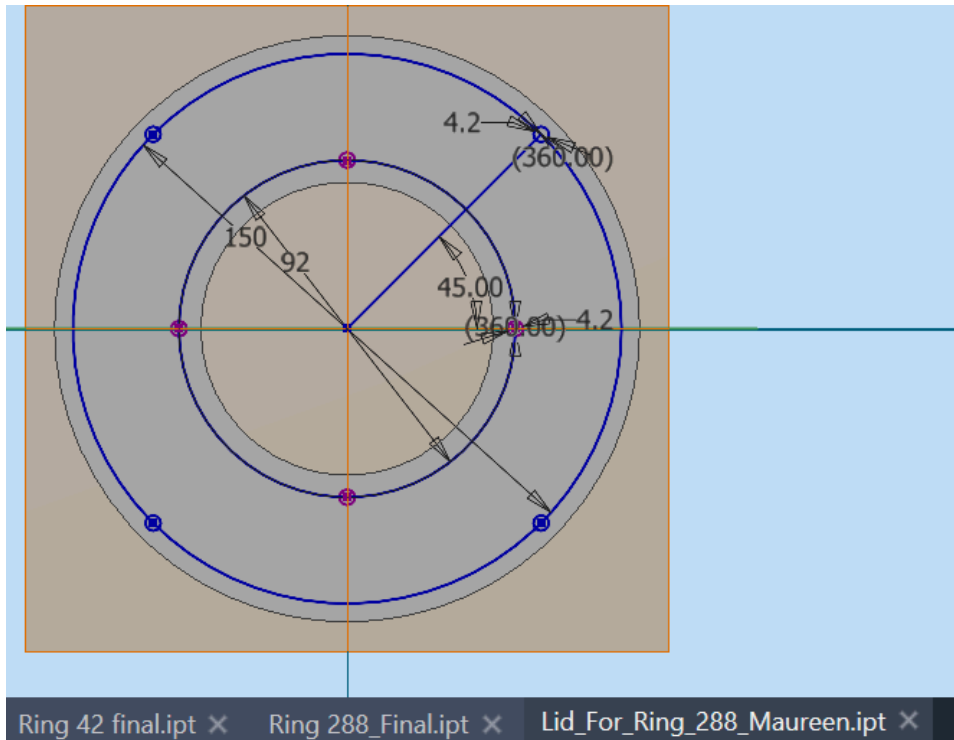
Ring 321

This is also different because the rings are more towards the exterior, so the diameter of the outer circle right after the band is 154mm to accommodate holes for screws.

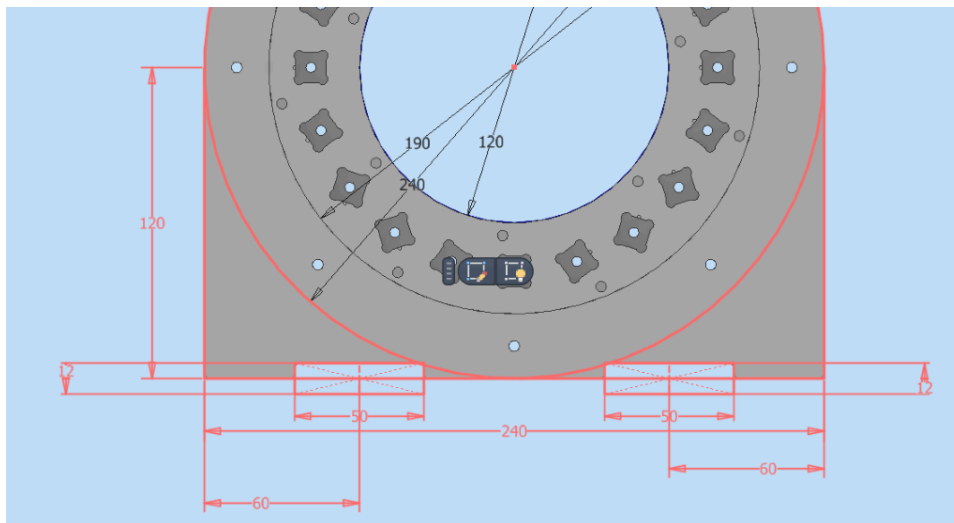
So the corresponding lid will be the same.

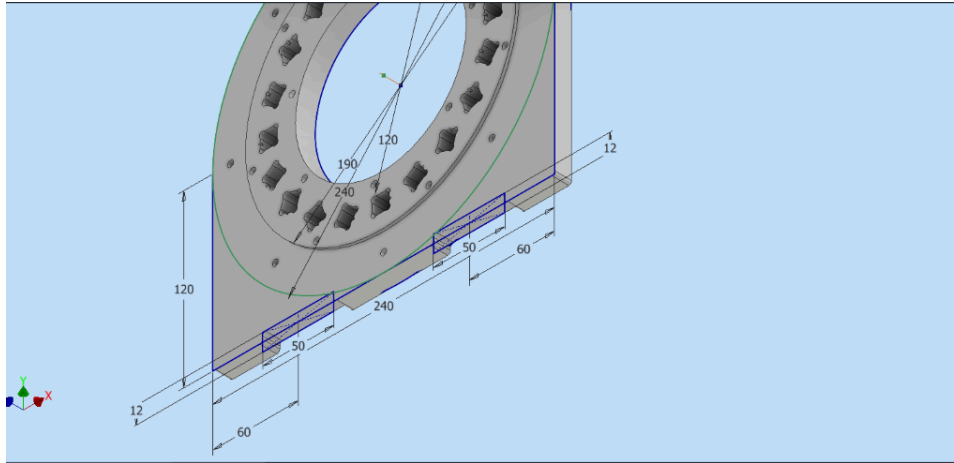
Lids

Lids For 288 and 251

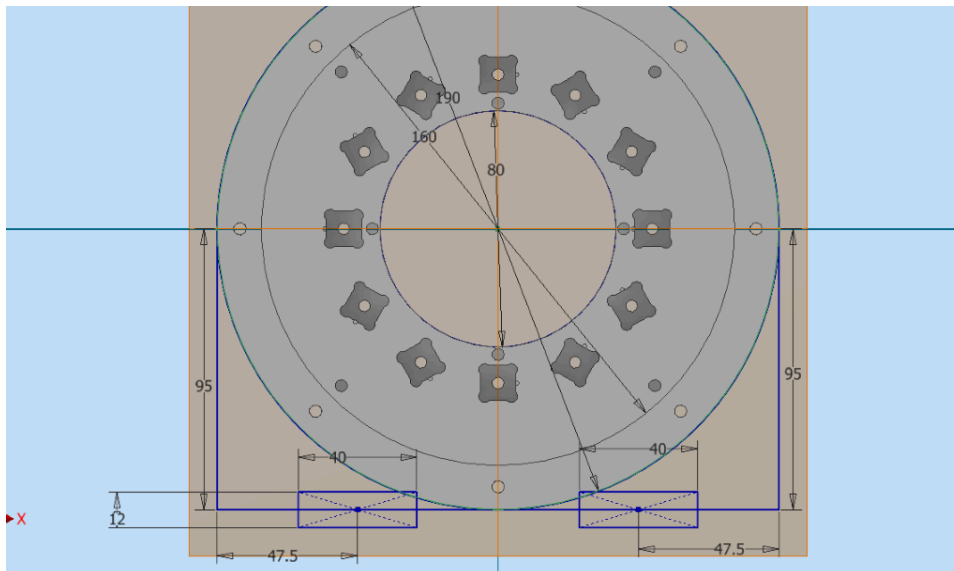


New sketch

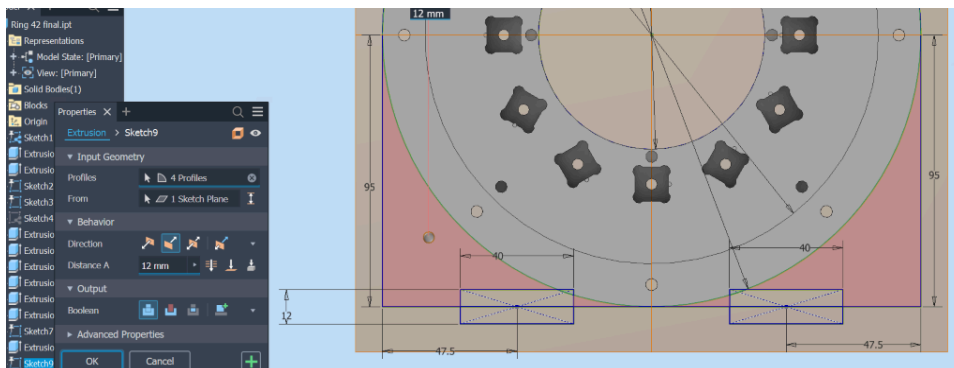




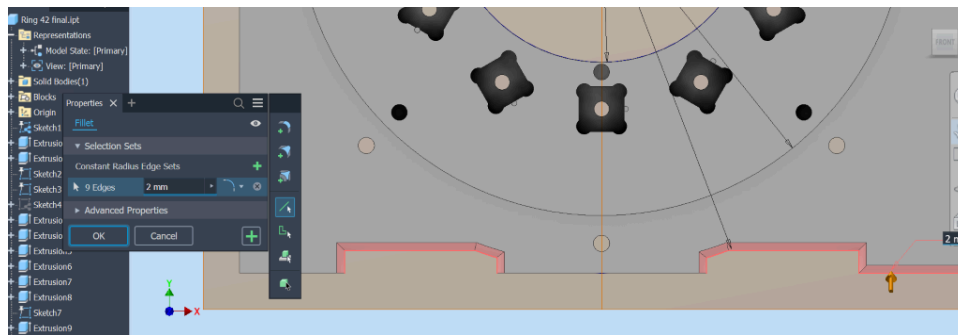
Since it is being scaled down, we set the specific dimensions in comparison to the 120mm diameter rat brain scanner



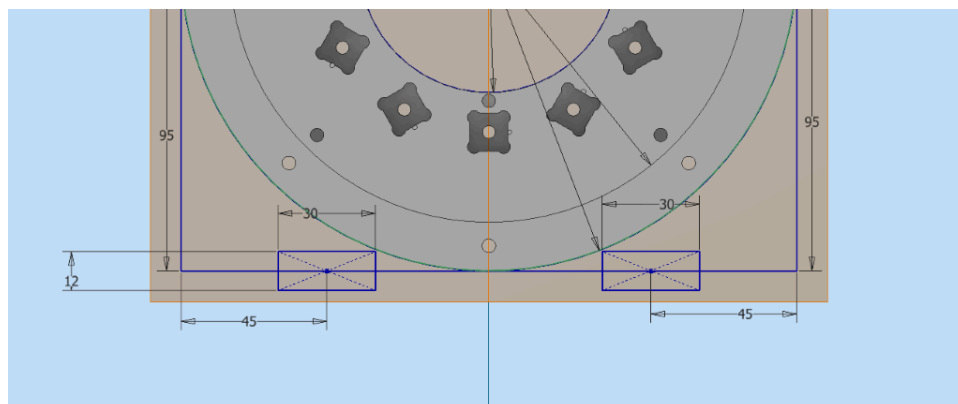
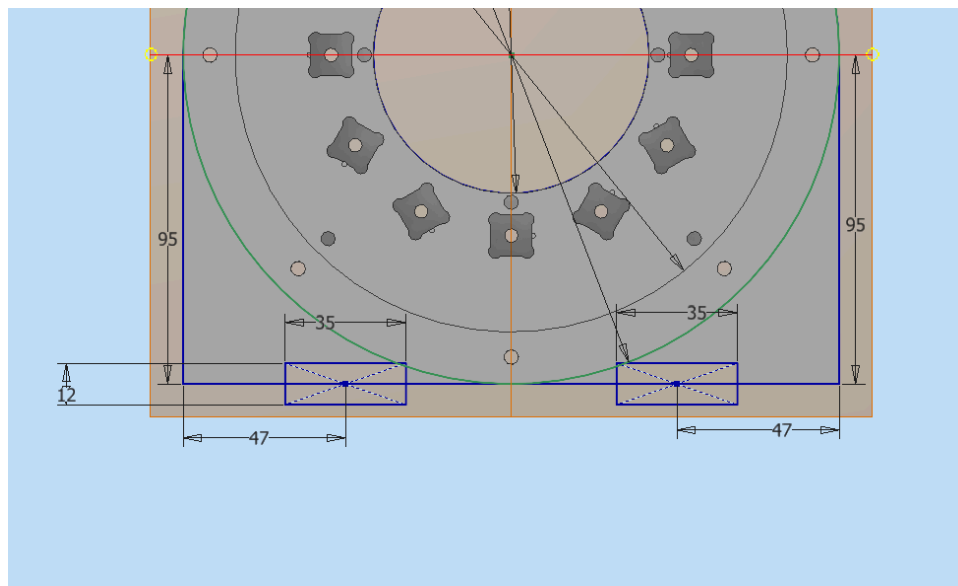
Then, extrude the support by 12mm



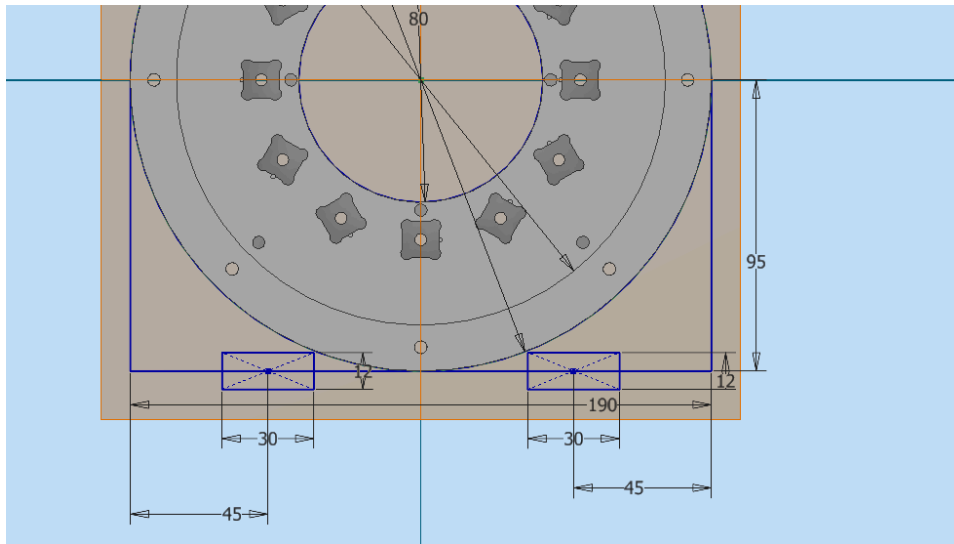
Fillet them



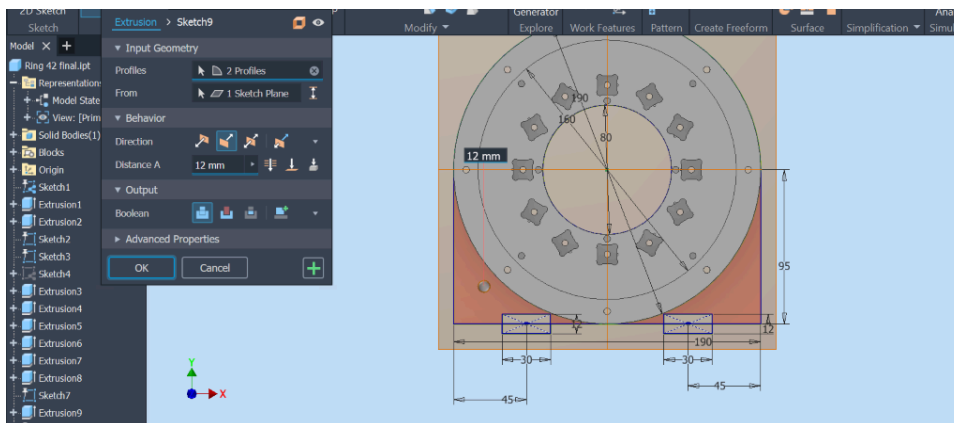
Try using different parameters



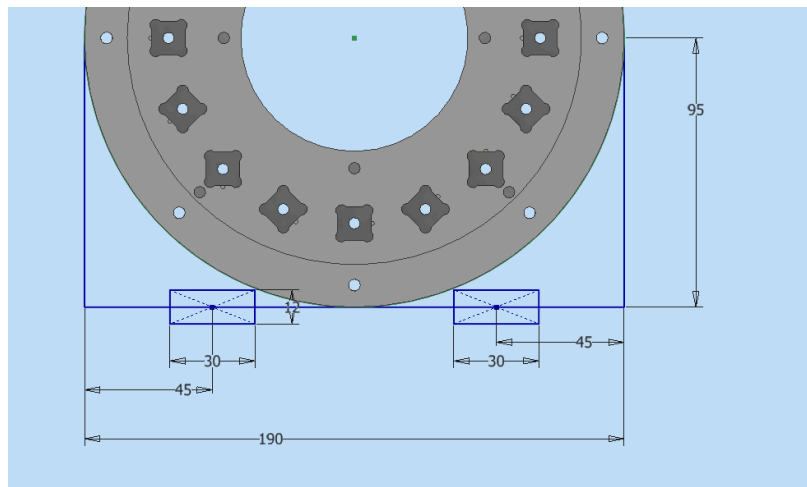
Final design



Extrusion



Ring 321



Fillet the edges

